

Book Recommendations at Scale: A Hybrid Neural Approach on the Goodreads Dataset

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Introduction

Over 129 million unique books have been published worldwide, far exceeding what any individual reader could explore. Identifying the right next book is therefore a fundamental recommendation problem. We present an end-to-end pipeline on the **full Goodreads dataset without subsampling** — from sentiment-based global ranking to a personalized hybrid neural network.

Input.

- User reviews (review text, rating, read_at)
- Book metadata (genre, language, year, authors)

Output.

- Global Top-N recommendation list (sentiment-based, non-personalized)
- Personalized Top-N recommendation list (hybrid neural network)

Key contributions.

- Full 15.7M review pipeline, no subsampling
- Hybrid Neural Network: CF embeddings + Content Features + Author embeddings + Two-phase Transfer Learning
- LLM-as-a-Judge evaluation with user segmentation analysis
- Statistically validated findings via paired t-tests
- Custom PyTorch — no off-the-shelf RecSys libraries

Dataset and Features

Dataset. UCSD Goodreads dataset (Wan et al., 2019):

15.7M Reviews	2.36M Books	465K Users
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Filter. Users ≥ 10 reviews, Books ≥ 25 reviews: 9,044,974 Reviews, 190,502 Users, 89,424 Books

Train/Validation/Test Split. Per-user chronological split on `read_at` field, eliminates cold-start

Train 70%	Val 15%	Test 15%
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Features.

- User (1):** Learned embedding $\mathbf{e}_u \in \mathbb{R}^{64}$ from rating history
- Book (1):** Learned embedding $\mathbf{e}_i \in \mathbb{R}^{64}$ from rating history
- Author (1):** Learned embedding $\mathbf{e}_a \in \mathbb{R}^{64}$
- Content (20):** Genre (10) + Language (6) + Numerical (4)
- Numerical:** Avg. VADER Sentiment Score, Mean Rating, Review Count, Published Year

Methods

Global Top-N (Sentiment Baseline).

Ranks books by vote-weighted average VADER [6] sentiment score (min 50 reviews).

Neural Collaborative Filtering.

Following He et al. [1]:

$$\hat{r}_{ui} = f(\mathbf{e}_u, \mathbf{e}_i; \Theta), \quad \mathbf{e}_u, \mathbf{e}_i \in \mathbb{R}^{64} \tag{1}$$

- MLP: $[128 \rightarrow 64 \rightarrow 1]$, ReLU, dropout $p=0.35$, Kaiming init [8] & $N(0, 0.01)$
- Adam [7] $\text{lr} = 0.001$, batch= 2048, ReduceLROnPlateau + early stopping (patience=7)

Methods (continued)

Hybrid Neural Network.

Extends NCF with author embeddings $\mathbf{e}_a \in \mathbb{R}^{64}$, 20-dim content features $\mathbf{c}$, and element-wise interaction term $\mathbf{z}_{ui} = \mathbf{e}_u \odot \mathbf{e}_i$:

$$\mathbf{x} = [\mathbf{e}_u || \mathbf{e}_i || \mathbf{e}_u \odot \mathbf{e}_i || \mathbf{e}_a || \mathbf{c}] \in \mathbb{R}^{276} \tag{2}$$

MLP Architecture: $[276 \rightarrow 128 \rightarrow 64 \rightarrow 1]$.

For multi-author books, author embeddings are combined using masked mean

Two-Phase Transfer Learning.

Randomly initialized MLP and author embedding layers produce large early gradients that, via backpropagation, corrupt pretrained NCF embeddings—reflected in the best validation RMSE at epoch 1 with subsequent degradation.

- Phase 1 (5 epochs):** Freeze $\mathbf{e}_u, \mathbf{e}_i \rightarrow$ stabilize author embeddings + MLP layers first
- Phase 2 (≤ 50 epochs):** Unfreeze all $\rightarrow$ fine-tune jointly at $\text{lr} = 0.0001$, allowing pretrained embeddings to adapt gently

End-to-End Pipeline.

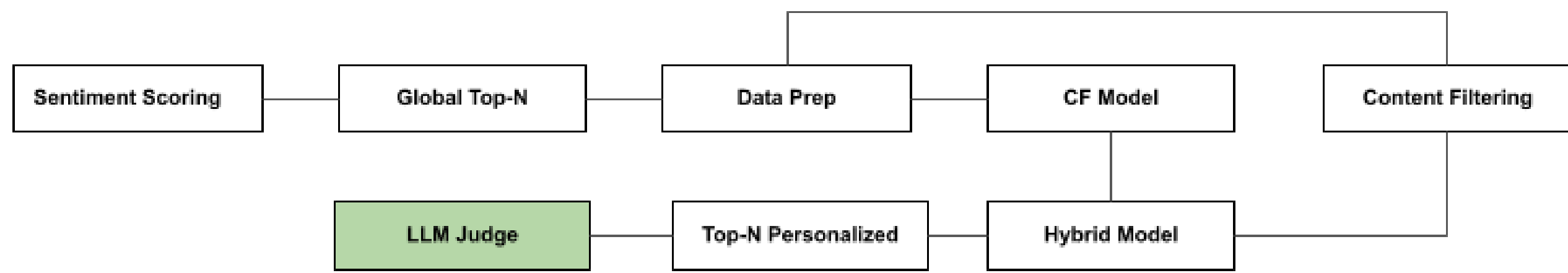


Figure 1. End-to-end recommendation pipeline.

Experiments / Results / Discussion

Evaluation Metric. $\text{RMSE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{(u,i) \in \mathcal{T}} (\hat{r}_{ui} - r_{ui})^2}$

Ablation Study — 7 Model Iterations.

Model / Configuration	Train	Val	Test
CF — no scheduler, dropout=0.20	0.9655	1.0919	1.2867
CF — scheduler, dropout=0.20	0.9792	1.0902	1.2847
CF — scheduler, dropout=0.35 ★	0.9803	1.0875	1.2803
Hybrid — single phase	1.1137	1.0894	1.2830
Hybrid — two phase	0.9948	1.0911	1.2833
Hybrid — two phase + interaction ★	1.0317	1.0900	1.2810
Hybrid — two phase + inter. + author init	0.9944	1.0930	1.2853

★ = best models selected for Top-N personalized prediction and LLM evaluation. Train RMSE at best val epoch.

- CF improvements.** Scheduler improved val RMSE; increasing dropout further helped
- Hybrid vs CF. Hybrid did not beat CF in RMSE.** With 9M interactions, CF embeddings likely already encode author/genre signals, consistent with Zhang et al [4].
- Two-phase + interaction.** Two-phase training stabilized gradients; explicit $\{\mathbf{e}_u \odot \mathbf{e}_i\}$ interaction yielded best hybrid RMSE
- Overfitting vs drift.** Train $\rightarrow$ val gap ~ 0.11 suggests **mild overfitting** (controlled by dropout + early stopping); larger val $\rightarrow$ test gap ~ 0.19 reflects **temporal preference drift**.
- Beyond RMSE:** Top-10 lists overlap only **4.87/10 books** — motivates LLM-as-a-Judge qualitative evaluation.

LLM-as-a-Judge Evaluation

To complement RMSE, we use an LLM-as-a-Judge framework [3].

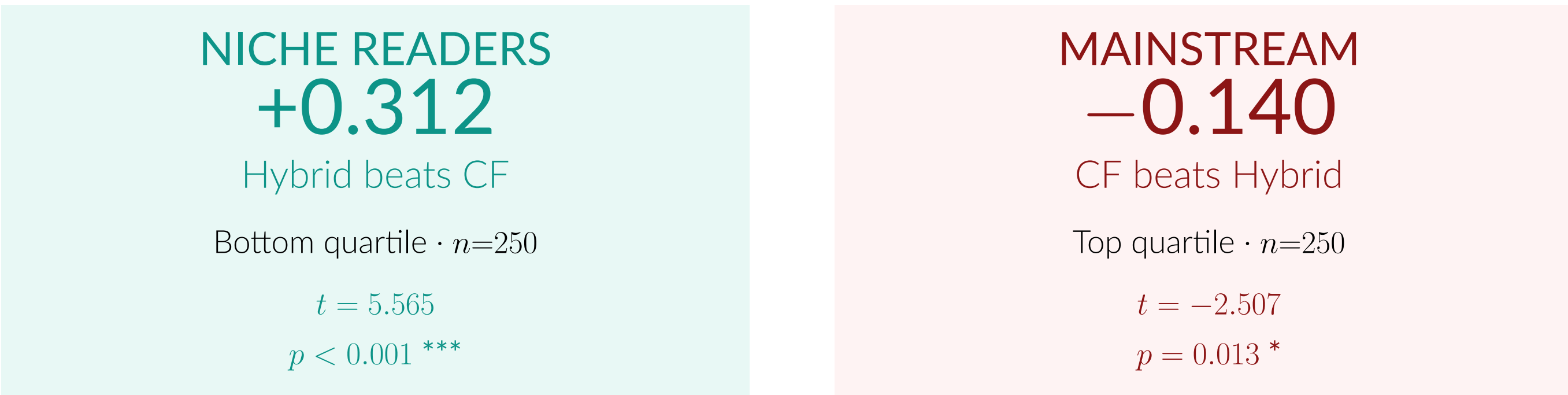
- Top 1,000 most active users (≥ 100 ratings, rating std ≥ 1.25).
- Claude scores Top-10 recommendations (1–10) given each user’s top-10 and bottom-10 rated books with no intersection between the lists.

Model	Mean Relevance (1–10)
CF (scheduler, dropout=0.35)	3.325
Hybrid (two phase + interaction)	3.447

- Hybrid outperforms CF by +0.122.**
- Absolute scores are low but the *relative difference* is the primary signal of interest.

User Segmentation Analysis — Key Finding

- Users segmented by avg review count of books** they rated – a proxy for niche vs. mainstream
- Statistical significance via paired t-test: $t = \bar{d} / (s_d / \sqrt{n})$, $H_0 : \bar{d} = 0$.



User Read Segment	N	CF	Hybrid	Relevance Δ	t-stat	p-value
Niche (bottom quartile)	250	2.668	2.980	+0.312	5.565	<0.001 ***
Mainstream (top quartile)	250	3.748	3.608	−0.140	-2.507	0.013 *
Overall	1,000	3.325	3.447	+0.122	4.103	<0.001 ***

Key Insights

- The observation that Hybrid outperforms CF for niche readers is **statistically verified**
- The observation that CF outperforms Hybrid for mainstream readers is **statistically verified**
- Content Features provides strongest signal **where CF is weakest** — sparse interactions

Conclusion & Future Work

- Best RMSE.** CF:{Validation: 1.0875, Test: 1.2803} **Hybrid**:{Validation: 1.0900, Test: 1.2810}
- CF embeddings capture most author/genre signal — content adds little RMSE gain at scale
- Hybrid wins for niche readers($p < 0.001$), CF wins for mainstream($p = 0.013$) – statistically verified
- Future work:** Productionize with online feedback learning, cold-start support, and scalable multi-tenant inference. Explore time-aware embeddings to reduce temporal drift and better align RMSE with real-world recommendation quality.

References

[1] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T.-S. Chua, "Neural collaborative filtering," in *Proceedings of the 26th International Conference on World Wide Web*, pp. 173–182, 2017.

[2] L. Taycher, "Books of the world, stand up and be counted! all 129,864,880 of you," *Google Books Search*, 2010.

[3] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, et al., "Judging llm-as-a-judge with mt-bench and chatbot arena," in *Advances in Neural Information Processing Systems*, 2023.

[4] S. Zhang, L. Yao, A. Sun, and Y. Tay, "Deep learning based recommender system: A survey and new perspectives," *ACM Computing Surveys*, vol. 52, no. 1, pp. 1–38, 2019.

[5] M. Wan, R. Misra, N. Nakashole, and J. McAuley, "Fine-grained spoiler detection from large-scale review corpora," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 2605–2610, 2019.

[6] C. Hutto and E. Gilbert, "Vader: A parsimonious rule-based model for sentiment analysis of social media text," in *Proceedings of the International AAAI Conference on Web and Social Media*, vol. 8, pp. 216–225, 2014.

[7] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[8] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on imagenet classification," in *Proceedings of the IEEE International Conference on Computer Vision*, pp. 1026–1034, 2015.